

JEE MAIN REVISION MASTER: D AND F BLOCK ELEMENTS 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	D And F Block Elements — Standard Calculations & Problem Archetypes	17 PYQs	Quantitative Analysis
High	D And F Block Elements — Applied Cases, Reactions & Traps	12 PYQs	Applied Topics
Moderate	D And F Block Elements — Core Principles & Governing Laws	4 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
D And F Block Elements — Standard Calculations & Problem Archetypes	$[Ar]3d^{10}4s^1$, $(n-1)d^{10}ns^2$, $\mu = \sqrt{n(n+2)} \text{ BM}$
D And F Block Elements — Applied Cases, Reactions & Traps	d^0 , Mn^{2+} , $2x-14=0$
D And F Block Elements — Core Principles & Governing Laws	Mn^{2+}

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 D And F Block Elements — Standard Calculations & Problem Archetypes (Very High Yield — 17 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in D And F Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The calculated spin only magnetic moment of Cr^{2+} ion is..." [Ans: B]
- [2025 Shift] Q#1 (MCQ): "A member of the Lanthanoid series which is well known to exhibit +4 oxidation state is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "In which of the following pairs, both the elements do not have $(n-1)d^1 0 ns^2$ configuration?..." [Ans: N/A]

3.2 D And F Block Elements — Applied Cases, Reactions & Traps (High Yield — 12 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of D And F Block Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "The highest oxidation state of manganese in fluoride is +4 (MnF_4), but the highest oxidation state in oxides is +7 (\$\text{Mn}_2\text{O}_7\$)..." [Ans: D]
- [2026 Shift] Q#3 (MCQ): "Which of the following will not act as an oxidising agent?..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "In the following pairs, the one in which both transition metal ions are colourless is..." [Ans: N/A]

3.3 D And F Block Elements — Core Principles & Governing Laws (Moderate Yield — 4 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in D And F Block Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in D And F Block Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in D And F Block Elements PYQs.

Representative Source PYQs:

- [2024 Shift] Q#3 (MCQ): " MnO exhibits..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "Which of the following is correct with respect to melting point of a transition element?..." [Ans: N/A]
- [2021 Shift] Q#3 (MCQ): "Which of the following does not represent property stated against it?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The calculated spin only magnetic moment of Cr^{2+} ion is

- (A) 3.87 BM
- (B) 4.90 BM **[Correct]**
- (C) 5.92 BM
- (D) 2.84 BM

Correct Answer: B

Solution Analysis: For spin only magnetic moment, we use the formula $\mu = \sqrt{n(n+2)}\text{BM}$ where n is the number of unpaired electrons. Chromium has atomic number 24. Its electronic configuration is $\text{Cr} = [\text{Ar}]3d^5 4s^1$. For Cr^{2+} , two electrons are removed. First the 4s electron is removed, then one electron from 3d. So, $\text{Cr}^{2+} = [\text{Ar}]3d^4$. In $3d^4$, according to Hund's rule, the four electrons occupy four different d orbitals singly. So the number of unpaired electrons is $n=4$. Now, $\mu = \sqrt{4(4+2)} = \sqrt{24}$ $\mu = 4.90\text{BM}$. So the correct answer is B.

Benchmark Problem 2: KCET 2026 (Q#2)

The highest oxidation state of manganese in fluoride is +4 (MnF_4), but the highest oxidation state in oxides is +7 (Mn_2O_7), because

- (A) Fluorine is more electronegative than oxygen
- (B) Fluorine possesses d-orbitals
- (C) Fluorine stabilises lower oxidation state
- (D) In covalent compounds, fluorine can form single bond only, while oxygen forms double bond **[Correct]**

Correct Answer: D

Solution Analysis: The correct option is Option D. In MnF_4 , fluorine has oxidation state -1. So, if oxidation state of Mn is x : $x + 4(-1) = 0$ $x = +4$. So manganese shows +4 oxidation state in MnF_4 . In Mn_2O_7 , oxygen has oxidation state -2. Let oxidation state of Mn be x : $2x + 7(-2) = 0$ $2x - 14 = 0$ $x = +7$. So manganese shows +7 oxidation state in Mn_2O_7 . Now the main reason is bonding. Fluorine is very electronegative, but it can form only one single bond because it needs only one electron to complete its octet. So fluorine cannot form double bonds.

Benchmark Problem 3: KCET 2024 (Q#3)

MnO exhibits

- (A) ferrimagnetism
- (B) antiferromagnetism
- (C) ferromagnetism
- (D) paramagnetism

Correct Answer: None

Solution Analysis: MnO exhibits antiferromagnetism. They have domain structure similar to ferromagnetic substance, but their domains are oppositely oriented and cancel out each other's magnetic moment.